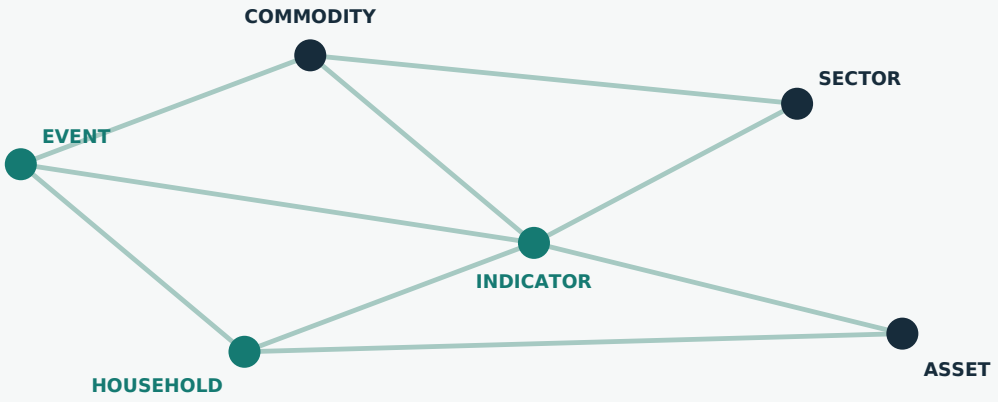
HACKWAVE 3.0 / PROBLEM STATEMENT 5

India Impact Graph

MVP Definition Document

From Global News to Your Household Budget



An evidence-first map of how financial events may travel through the economy

The product in one sentence

An evidence-first application that turns a financial news event into an explorable impact graph, shows supporting sources for important links and numbers, and estimates a typical Indian household's monthly spending impact only where the required data is available.

Version: 1.0 **Date:** 10 October 2026
Document type: Product and MVP definition - not implementation code

Document map

Use this document as the shared agreement for what the MVP must do, what it must not do, and how the team will decide that it is ready.

01 Executive summary and problem
02 Product goal, users, and USP
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12 Glossary and source basis

Non-negotiable data rule

The AI may read and explain. It must not invent prices, expenditure weights, pass-through factors, sources, or rupee-impact values. Calculations must use explicit inputs and deterministic formulas. When required data is missing, show “Data not available”.

Status labels used in this document

Label	Meaning
[PS] Official requirement	Explicitly described in the supplied Problem Statement.
[DECISION] Product choice	The team's selected MVP behaviour or scope.
[VERIFY] Before use	A dataset, exact table, relationship, value, access condition, or source must be checked.
[INFERENCE]	A plausible relationship or explanation that is not a directly observed fact.
[OUT OF SCOPE]	Not part of the MVP unless the team explicitly changes scope.

01. Executive summary and problem

Official Problem Statement: “Autonomous Financial Event Market Impact Graph.”

Financial events rarely affect only one company. A geopolitical event, commodity shock, interest-rate decision, or regulatory change may propagate through sectors and financial instruments. Many systems present news as separate stories instead of connected chains of impact. The challenge is to convert unstructured financial information into a dynamic dependency graph and explain how an event may influence economic variables, sectors, and assets.

Simple Hinglish summary

News alag-alag dikh sakti hai, lekin ek financial event ka asar kai connected steps se aage ja sakta hai. India Impact Graph un steps, unke evidence, uncertainty, aur jahan data available ho wahan typical Indian household ke monthly budget par approximate impact ko ek jagah dikhata hai.

What the official PS expects [PS]

- Detect financial events from unstructured text.
- Extract companies, commodities, sectors, indicators, and financial instruments.
- Identify and score relationships between entities.
- Construct and update a knowledge graph.
- Estimate plausible propagation paths and affected assets.
- Use time-series data where available.
- Explain each impact chain and the evidence behind it.
- Distinguish observed facts from inferred relationships and communicate uncertainty rather than guarantees.

Our product interpretation [DECISION]

Build an India-focused experience that starts with a financial news article or a clearly labelled shock scenario, draws the relevant impact graph, lets the user inspect evidence behind important relationships, and presents an estimated monthly household spending impact in rupees only where the input data supports a defensible calculation.

Success does not mean predicting the market correctly. It means making the event, relationships, evidence, uncertainty, and calculation path inspectable and testable.

02. Product goal, users, and USP

Product name: India Impact Graph

Tagline: **From Global News to Your Household Budget**

Primary users are not specified in the official PS. For this MVP, the intended users are hackathon judges and people who want to understand how a financial event may flow through the economy to household expenditure. This is a team product decision, not an official requirement.

Our USP: Evidence-First Financial Impact Graph

Every important impact connection should show why it exists, what evidence supports it, whether it is observed or inferred, and what could weaken the relationship. The product also connects the graph to a household persona impact card, with number-level source receipts.

Differentiator	What the user sees	Why it matters
Receipt-backed numbers	Click a number to see source, table/series, year, unit, and formula where relevant.	Users can inspect where a number came from; unsupported values are not hidden.
Persona Impact Card	Estimated monthly impact range in rupees for state + rural/urban + expenditure class.	Turns abstract percentages into a household-oriented explanation, without claiming individual-level precision.
News Consensus Meter (P1)	Comparison of up to three saved sources, agreement, conflicts, and source links.	Makes source disagreement visible; consensus is not treated as proof.
AI-off proof (P1)	Disable the LLM while curated graph paths and deterministic calculations remain usable.	Shows that the AI explains text but does not secretly calculate the numbers.

Pitch line

From global news to your household budget. Every number comes with a source receipt, and AI reads and explains - it never does the maths.

Product principles

- Evidence before decoration: the graph must be useful even if the visual styling is basic.
- Observed facts and inferred links must look different and remain different in the stored data.
- Household impact is an estimate range, not a guarantee or a personal forecast.
- Missing data is a valid output state, not a reason to invent a value.
- Keep the first working end-to-end flow small before adding optional analytics.

03. MVP scope and exclusions

Scope decisions [DECISION]

Area	Included in MVP	Boundary / rule
Market	India-focused analysis with World/India/State/Persona views.	Do not claim city-level or individual household results.
States	Madhya Pradesh plus up to two more states if official data is verified.	If data is unavailable, show the status; do not fill gaps.
Persona	State, rural/urban, low/middle/high expenditure class.	Use “expenditure class”, not “income”. Personas represent groups, not real people.
Items	8-10 candidate items: petrol, diesel, LPG, edible oil, cereals, pulses, sugar, vegetables, electricity, gold.	Only calculate an item when its spending share and relevant price/pass-through inputs are supported.
Events	Primary crude oil +12% scenario; one backup edible-oil or wheat event if reliable data is available.	A +12% shock is illustrative unless a matching historical event and dates are verified.
News	Three cached articles/source documents per event where available.	No live news stream required. Show the actual source count.
Household size	Use official household-size data only if needed for household-level conversion.	Otherwise label result as per-capita or disclose the limitation.
CPI series	Use the official applicable series and metadata.	Do not silently mix 2012-base and 2024-base CPI series.

Explicitly out of scope [OUT OF SCOPE]

- Live news streaming, continuous scraping, or an always-on alerting service.
- User accounts, login, personal financial profiling, or real-person data.
- City-level claims, more than five states, or unsupported granular household claims.
- Brokerage or trading integration, buy/sell recommendations, or stock-price prediction models.
- Custom-trained LLMs, multi-agent systems, complex market simulation, or unverified backtesting.
- Advanced export and dark mode before core functionality is complete.

MVP boundary

The system explains plausible impact pathways. It does not promise that a price will move, calculate guaranteed financial losses, or claim that a single event caused every downstream market movement.

04. User journey and system behaviour

The first version must support one complete flow from a news input to an explorable graph and a transparent household-impact result.

1. Select or paste event

Choose the primary crude-oil +12% scenario, select a saved article, or paste a news article. Show known source/date information. Mark the +12% scenario as hypothetical unless verified as a specific historical observation.

2. Extract event details

The LLM reads the supplied text and returns validated structured fields: event type, magnitude/unit when stated, date, geography, entities, and evidence snippets. Missing fields remain missing.

3. Match the curated graph

Map extracted entities to the source-backed base graph. The LLM may not add an unsupported edge and call it a fact.

4. Explore impact paths

Show the connected graph and three proposed paths where evidence supports them. Users can highlight a path and click a relationship.

5. Inspect evidence

An edge panel explains the relationship, status, source, date, metric/table, unit, evidence quality, assumptions, and what could be wrong.

6. Select a household persona

Choose state, rural/urban, and expenditure class. Show which official expenditure data is available for this combination.

7. Review impact range

Calculate an estimated monthly rupee range only for eligible items. Show the inputs, formula, data coverage, assumptions, exclusions, and receipts.

8. Compare sources / data

When P1 is stable, inspect the time-series chart, News Consensus Meter, and AI-off toggle. Numeric results must not change when the LLM is disabled.

High-level processing flow

Stage	Responsibility	Output
Input	User + cached article/scenario	Text and known metadata
Reader	LLM + schema validation	Structured event/entity JSON
Graph matching	Curated, sourced relationships	Nodes and typed edges
Path finder	Deterministic graph algorithm	Ranked plausible paths
Impact calculator	Formula + sourced CSV/JSON data	Low-to-high estimate where support exists
Explainer	LLM constrained to supplied evidence	Plain-language explanation and caveats
UI	Interactive graph, receipts, persona card	Inspectible result and clear missing-data states

05. Knowledge graph and propagation paths

A knowledge graph is a network of entities (nodes) connected by relationships (edges). A propagation path is a route through that network showing how a possible effect may travel from an event to other entities.

Required node types

Node type	Examples in this MVP
Commodity	Crude oil, petrol, diesel, edible oil, wheat/cereals
Household item	Food, cooking fuel, commuting-related expenditure, electricity
Economic indicator	CPI / inflation indicator, relevant price series
Sector	Transport, food supply/logistics, banking, real estate
Instrument / asset	Relevant market index or asset, only when relevant evidence exists
Persona	State + rural/urban + expenditure class

Edge rules

- Observed edge: directly stated in a source or directly measured by an identified dataset; show as a solid line.
- Inferred edge: a plausible or estimated link that is not directly observed in the supplied evidence; show as a dashed line.
- Every important edge must carry a relation type, direction where supported, evidence, source, year/date, and uncertainty label.
- If the evidence does not support a link, do not force the path to complete. Show “unverified” or “Data not available”.
- Line thickness represents strength only if there is a defensible, documented value. Otherwise keep line width constant.
- An LLM explanation is not independent proof of a relationship.

Three proposed propagation paths

Path	Proposed chain	Evidence caveat
A. Household food-cost	Crude oil → Diesel → Freight costs → Food/vegetables → Household budget	Verify each edge separately. Freight-to-retail pass-through depends on contracts, time, margins, and local supply.
B. Household transport	Crude oil → Petrol → Commuting cost → Household budget	Do not assume domestic retail petrol prices move one-for-one with global crude oil.
C. Financial market	Crude oil → Inflation expectations → RBI-rate expectations → Banks/real estate → Relevant indices	These are conditional relationships. The chain must not imply that every step is observed or that an asset-price move is guaranteed.

Historical case versus scenario

The project brief defines “crude oil +12%” as its main demo scenario. A scenario input is not proof that a real historical event had exactly that magnitude. The app must label scenario assumptions separately from dated historical observations.

06. Household impact calculator and data rules

The Persona Impact Card is a core differentiator. It translates supported item-level price changes into an approximate monthly rupee impact for a typical expenditure persona. It must never imply an exact prediction of a real person's future household bill.

Required inputs

Input	Definition	Rule
Persona MPCE	Official monthly per-capita consumption expenditure for the chosen group.	Source table, year, geography, rural/urban and expenditure class must be recorded.
Item spending share	Share of expenditure attributed to a selected item/group.	Use applicable official HCES/CPI tables; exact table and item mapping must be verified.
Event shock	Magnitude from article or selected scenario, e.g. +12% crude oil.	Source it or label it scenario input.
Pass-through factor	Estimated share of a global price change reflected in an Indian retail price.	Must be supported by historical data or a sourced, visibly labelled estimate.
Household-size conversion	Only used if converting MPCE to a household total.	Use verified source data; otherwise label output per-capita or disclose limitation.

Formula

Deterministic calculation

Item spending = Persona MPCE × item expenditure share

Item impact = Item spending × item price-change percentage × pass-through factor

Total impact range = Sum of eligible item impact ranges

Low and high results must use explicit lower and upper assumptions. Apply pass-through once only; never have the LLM calculate or adjust numeric outputs.

Calculation and display rules

- A +12% crude-oil shock does not mean petrol, diesel, LPG, food, or retail prices also increase by 12%.
- Estimate pass-through for only the items where a reliable historical basis can be obtained. The project brief proposes prioritizing roughly 3-4 items.
- For all unsupported items, display "Data not available"; do not replace missing values with zero or AI-generated estimates.
- Always show low-to-high range, item coverage, low/high assumptions, units, source receipts, and calculation steps.
- Keep household reach (how many households use an item) separate from monetary impact (how much the budget is affected).
- If only per-capita expenditure is supported, label the output per capita. Do not silently multiply by an invented household size.
- An estimated rupee range is a scenario analysis, not financial advice or a guarantee.

CPI series integrity

The project brief calls out the transition from the 2012-base CPI series to a 2024-base series. The precise official series, weights, linking factors and back-series documents must be checked before use. Do not mix series directly or imply comparability unless the official linking methodology has been correctly applied and labelled.

07. Data sources and evidence receipts

The sources below are candidates from the project brief. This document does not claim that each exact table, API, permission, pass-through series, or item weight has already been downloaded and validated. Items marked [VERIFY] must be confirmed before they feed a calculation.

Data need	Candidate source	Required check / status
CPI item weights	MoSPI CPI (2024 base)	[VERIFY] Exact official weight table, item mapping, year and applicable series.
Spending by state / rural-urban / expenditure class	MoSPI HCES 2023-24	[VERIFY] Exact tables, MPCE values, class definitions and item-group shares.
Global commodity prices	World Bank Pink Sheet, FRED, FAO where appropriate	[VERIFY] Series suitability, units, dates, download/API conditions.
Indian fuel prices	PPAC	[VERIFY] Exact retail price series, geography and historical availability.
Indian retail price history	MoSPI / RBI, as applicable	[VERIFY] Item, geographic coverage, frequency and series definition.
Historical financial market series	FRED / suitable official market source; yfinance only if acceptable for prototype	[VERIFY] Data terms, ticker coverage, units, dates and reliability.
Shipping freight rates, if used	Freightos / Drewry	Potentially paid or restricted; do not make MVP depend on access.
News evidence	Three saved articles/source documents per event where available	Record title, publisher, publication date, event date, direct URL and evidence excerpt.

Every number: the “receipt” fields

- Metric or item name, value, unit, date/year and geography.
- Source title and direct URL; table, series or document identifier.
- Data base/version where relevant (especially CPI series); verification status.
- Formula and input values for derived numbers.
- Observed/inferred status and assumptions, where the number supports a relationship.
- If missing or unverified, say so explicitly. Never create a fake source receipt.

Three source types are not interchangeable

A news article can establish what was reported; a dataset can establish a measured value; a company filing or economic study may support a relationship mechanism. A time-series chart showing two variables moving together does not, on its own, prove causality.

08. Feature priorities and screen requirements

Prioritize the smallest complete workflow. P1 work starts only after P0 works end-to-end.

Priority	Feature	Definition of done
P0 - Must build	News/scenario input + extraction	Select/paste article; extract event, magnitude if stated, geography, entities and evidence into validated structured fields.
P0 - Must build	Interactive graph	Show relevant nodes and directional edges; solid observed, dashed inferred; users can inspect paths.
P0 - Must build	Three propagation paths	Provide the proposed paths where evidence supports them; incomplete paths clearly show missing/unverified edges.
P0 - Must build	Evidence receipts	Click an important edge or number to see source, date/year, unit, evidence, assumptions and formula as relevant.
P0 - Must build	Persona selector + impact card	Select state, rural/urban, and expenditure class; calculate low-high impact only for supported items.
P0 - Must build	Data integrity states	Missing/unverified data is visibly labelled; no hidden zeros or AI-invented amounts.
P1 - Differentiator	Time-series chart	Crude oil compared with one or two relevant Indian fuel/CPI series, with units, period, source and caveats.
P1 - Differentiator	AI-off toggle	LLM contribution disabled; same inputs yield identical deterministic calculations.
P1 - Differentiator	News Consensus Meter	Compare up to three saved sources, shared claims and conflicts; no fabricated sources.
P2 - If time	Historical backtest / second event / state comparison	Only with suitable data and clearly labelled series versions; one backup edible-oil or wheat event if supported.
P2 - If time	“What to watch” notes	Rule-based and sourced; not investment advice.
P3 - Polish	Export / dark mode	Only after core acceptance criteria and P1 stability.

Main screens / components

- Header: product name, event selector, AI status.
- Event input: saved article or pasted text, event metadata and source.
- Impact graph: nodes, solid/dashed edges, path highlighting, observed/inferred filters.
- Evidence panel: relationship reason, source excerpt, source/date, confidence/evidence quality, assumptions and limitations.
- Persona panel: state, rural/urban, expenditure class, verified coverage, rupee range and item-level breakdown.
- Supporting analysis: time-series chart, News Consensus Meter and AI-off toggle.

09. 36-hour build plan

This plan is a recommendation for sequencing work, not an official organiser schedule. Data-access checks should happen early because the MVP depends on verified tables and historical series.

Time	Goal	Deliverable / checkpoint
0-3 hours	Scope and data check	Confirm one event input, source documents, available official tables, missing [VERIFY] items, and expected graph paths. Download/check critical datasets early.
3-8 hours	Data foundation + extraction	Create structured data files with source/year/unit fields; implement event extraction to validated JSON; test on saved articles.
8-16 hours	P0 graph workflow	Construct curated graph, observed/inferred styling, path highlighting and evidence panel.
16-24 hours	Persona and calculator	Implement persona selectors, item spending, pass-through inputs, low/high calculation, item coverage and receipts.
24-30 hours	Reliability and integration	Connect UI to data and extraction; handle missing data/API errors; test every edge and calculated number.
30-34 hours	P1 only if P0 stable	Add time-series chart, AI-off proof, and News Consensus Meter if the data and time permit.
34-36 hours	Freeze and demo rehearsal	Freeze features, verify startup, test cached fallback, rehearse three-minute demo, prepare known limitations.

Suggested parallel responsibilities

Workstream	Accountability
Data / research	Official datasets, source receipts, item mapping, price series and verification log.
Extraction / graph	Structured event extraction, schema validation, curated edges, path-finding logic.
UI / interaction	Graph display, selectors, evidence panel, persona card and interaction states.
Integration / QA	End-to-end flow, formula tests, AI-off equality, fallback path, demo and documentation.

Critical path

Verified inputs → structured event extraction → curated graph → evidence-supported paths → deterministic persona calculation → receipts → three-minute demonstration.

First 12 hours: do not build

- A custom model, multi-agent architecture or stock-price predictor.
- Broad all-market/all-company coverage or a live news feed.
- Advanced animation, exports, login or dark mode.
- A complex quantitative simulation before the evidence-backed graph works.

10. Demo script and acceptance criteria

Target demo duration: approximately three minutes.

Time	Demo action	What it proves
0:00-0:20	Explain that news often appears as separate stories, not as connected impact chains.	Problem understanding.
0:20-0:45	Select “Crude oil +12%” and clarify whether it is a scenario or verified historical event.	Transparent input and event labelling.
0:45-1:15	Show extracted entities and the interactive graph; highlight three paths where evidence supports them.	Event extraction and graph/path functionality.
1:15-1:45	Click an edge to reveal evidence, source, dates, status, assumptions and uncertainty.	Evidence-first USP.
1:45-2:20	Select a rural Madhya Pradesh expenditure-class persona; show impact range only if the necessary data is verified. Click a number to view its receipt.	Household-impact differentiator and data integrity.
2:20-2:40	Switch AI off and show that deterministic values remain unchanged.	AI is reader/explainer, not the calculator.
2:40-3:00	Close: explain how far a possible effect can travel and how certain the evidence is.	Responsible interpretation, not a guarantee.

Acceptance checklist

- ☐ A user can select or paste a financial event.
- ☐ Event details are returned in validated structured fields; unknown fields remain unknown.
- ☐ Graph displays relevant entities and relationships with observed/inferred visual distinction.
- ☐ Three plausible paths can be explored where supporting evidence exists.
- ☐ Every important edge can be inspected and its source/evidence reviewed.
- ☐ Persona can be selected by state, rural/urban, and expenditure class.
- ☐ Rupee impact shows a low-to-high range only for supported inputs, with item-level coverage.
- ☐ Each displayed important number has a source receipt or transparent formula; missing values are not fabricated.
- ☐ AI-on and AI-off modes return identical deterministic calculations.
- ☐ App remains usable with saved articles/data if external services fail.
- ☐ Primary flow has been tested at desktop and mobile widths; no critical build errors remain.

If a required source is missing

The correct demo behaviour is to show “Data not available” and explain why the path or calculation is incomplete. Do not replace missing information with an invented number simply to make the graph look finished.

11. Risks, fallbacks, and open decisions

Risk	Why it matters	Fallback / mitigation
Pass-through estimates are noisy or unavailable	A crude-oil shock does not transfer one-to-one to domestic retail prices.	Calculate only 3-4 items if supported; otherwise mark item “Data not available”. Show low/high assumptions.
LLM invents an edge or amount	Could mislead the user and undermine trust.	Curated base graph, schema validation, fixed relation vocabulary, source evidence checks; unverified links stay inferred/low confidence.
API limits or internet failure	Live fetch could fail during demo.	Cache saved articles and verified CSV data; provide explicit offline/cached mode.
Official data lacks required granularity	False precision could be implied for state/item/household.	Use broad official item groups and correct geography; disclose limits; do not interpolate silently.
CPI series mismatch	2012-base and 2024-base series may not be directly comparable.	Use the applicable series and official linking methodology; label old series in a backtest.
News sources disagree	A consensus meter can overstate certainty.	Show claims side by side and state what conflicts; consensus is not independent proof.
Correlation described as causation	A chart may show co-movement for unrelated reasons.	Use observed/inferred labels, alternative explanations, and cautious wording.
Scope creep	Too many domains/features can prevent a working MVP.	No P1 until P0 is stable; add data-rich items only after core flow passes.

Open decisions the team must close

- Which two additional states beyond Madhya Pradesh have the necessary official tables?
- Which three cached source documents are used for each event, and what is each source verifying?
- What exact CPI 2024-base weights file and HCES item-group tables will be used?
- Which 3-4 item pass-through relationships can be supported with available historical data?
- Which backup event (edible oil or wheat) has the strongest verified data for an MVP?
- Who owns data verification, graph/extraction, UI, integration and final QA?
- Which values can honestly be shown as a monthly household estimate, and which must remain per-capita or unavailable?

Go / no-go rule

Do not call the MVP ready until the event-to-graph flow and the evidence panel work. Do not call the household calculator complete unless its rupee range can be reproduced from the cited input data and formula.

12. Glossary and source basis

Term	Simple meaning
Event extraction	Identifying what happened (e.g. a stated crude-oil price shock) from a news article.
Entity	A named thing such as a commodity, company, sector or economic indicator.
Knowledge graph	Nodes representing entities and edges representing relationships.
Propagation path	A route through the graph showing how a possible effect may travel across connected entities.
Observed vs inferred	Observed is directly stated or measured; inferred is a supported estimate or plausible relationship not directly observed.
CPI	Consumer Price Index, a measure of consumer-price changes.
HCES	Household Consumption Expenditure Survey, an official survey of household spending.
MPCE	Monthly Per Capita Consumption Expenditure.
Pass-through	How much of an upstream/global price change appears in a downstream/Indian price.
Persona	A representative expenditure group defined by state, rural/urban, and expenditure class.
Receipt	The source/table/year and, where relevant, formula shown behind a number.
Consensus meter	A comparison of what multiple sources report; agreement does not automatically prove truth.

Source basis and limitations

This MVP definition is based on the supplied official Problem Statement image, the supplied “India Impact Graph: Complete Project Brief” PDF (version 1.0, dated 10 October 2026), and the supporting financial-domain research report. The official PS defines required capabilities; the project brief defines team scope and data rules; the research report contributes candidate events, domains and data-source suggestions.

The data rows and source relationships required for production are not all included or verified in this document. Items explicitly marked [VERIFY] remain unresolved until the team confirms the source, exact table/series, date range, licensing/access, and applicability. This document defines the MVP requirements; it is not a claim that an application has already been built or that every dataset has been downloaded.

Final product rule

Final product rule

Evidence before confidence. Data before numbers. A working end-to-end MVP before optional features.

END OF MVP DEFINITION DOCUMENT